📘 **AI/ML Internship – Day 18 Notes & Tasks**

🟡 **Module 4: NumPy (Fundamentals – Part 1)**

📌 **Part 1: What is NumPy?**

🔹 **Definition**

NumPy (Numerical Python) is a powerful library used for:

- Numerical computations
- Working with arrays
- Fast data processing

🔹 **Why NumPy?**

👉 Problems with Python lists:

- Slow for large data
- Cannot perform vector operations easily

👉 NumPy advantages:

- Faster (optimized in C)
- Less memory usage
- Supports mathematical operations directly

📌 **Part 2: Installation & Import**

🔹 **Install NumPy**

```
pip install numpy
```

🔹 **Import NumPy**

```
import numpy as np
```

📌 **Part 3: Creating Arrays**

🔹 **1D Array**

```
arr = np.array([1, 2, 3, 4])
print(arr)
```

🔹 **2D Array**

```
arr2 = np.array([[1, 2], [3, 4]])
print(arr2)
```

🔹 **3D Array**

```
arr3 = np.array([[[1, 2], [3, 4]]])
print(arr3)
```

📌 **Part 4: Array Properties**

🔹 **Shape**

```
print(arr2.shape)
```

🔹 **Dimensions**

```
print(arr2.ndim)
```

🔹 **Data Type**

```
print(arr.dtype)
```

📌 **Part 5: Special Arrays**

🔹 **Zeros**

```
np.zeros((2, 3))
```

🔹 **Ones**

```
np.ones((2, 2))
```

🔹 **Identity Matrix**

```
np.eye(3)
```

🔹 **Range**

```
np.arange(1, 10)
```

🔹 **Random Numbers**

```
np.random.rand(3)
```

📌 **Part 6: Indexing & Slicing**

🔹 **1D Array**

```
arr = np.array([10, 20, 30, 40])

print(arr[0])
print(arr[1:3])
```

🔹 **2D Array**

```
arr2 = np.array([[1, 2], [3, 4]])

print(arr2[0, 1])
```

📌 **Part 7: Reshaping Arrays**

```
arr = np.array([1, 2, 3, 4, 5, 6])

reshaped = arr.reshape(2, 3)
print(reshaped)
```

📌 **Part 8: Real-World Thinking**

👉 In AI/ML:

- Data = arrays
- Images = matrices (2D arrays)
- Videos = 3D arrays

👉 NumPy is base for:

- Pandas
- Machine Learning
- Deep Learning

🎯 **Tasks for Day 18**

🧠 **Task 1: Theory**

1. What is NumPy?
2. Why NumPy is faster than lists?
3. What is array?
4. What is shape and dimension?
5. What is reshaping?

💻 **Task 2: Array Creation**

1. Create 1D array
2. Create 2D array
3. Create 3D array
4. Print all

💻 **Task 3: Properties**

1. Print shape
2. Print dimensions
3. Print data type

💻 **Task 4: Special Arrays**

1. Create zero matrix
2. Create ones matrix
3. Create identity matrix
4. Create range array

💻 **Task 5: Indexing**

1. Access first element
2. Slice array
3. Access element in 2D

💻 **Task 6: Reshaping**

1. Convert 1D → 2D
2. Convert 2D → 1D